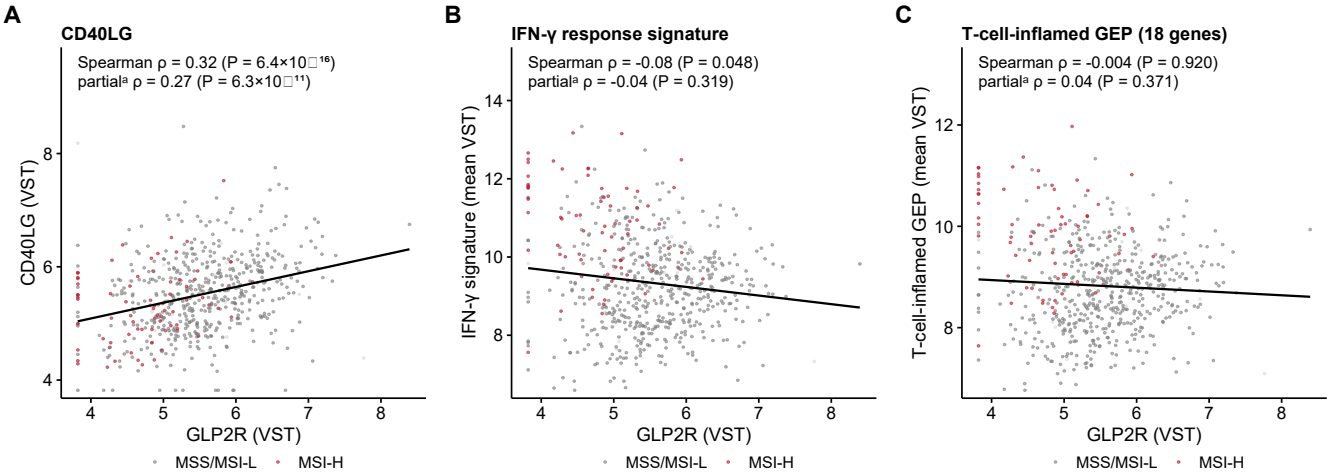


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Evidence summary

Question	Key result (TCGA-COADREAD unless stated)	Verdict
Tumour loss of expression	Lower in tumour vs normal: log□FC −4.07 (FDR 2×10□⁶⁶); paired 32/32 pairs lower, log□FC −3.82	Supported
MSI-H association	Lower in MSI-H than MSS/MSI-L (AUC 0.19; P 3.5×10□¹⁷)	Supported
Stage association	No association with stage (Spearman ρ 0.05; P 0.31)	Not supported
Prognostic association	Unadjusted OS: HR per SD 0.82 (0.68–0.98), P 0.032	Weak / descriptive
Independent prognostic value	After age, sex, stage, site, MSI: HR 0.86 (0.69–1.07), P 0.18	Not supported
External validation (microarray)	Adjusted HR 0.83 (TCGA), 1.00 (GSE39582), 0.71 (GSE17536); 3-cohort summary 0.86 (0.70–1.04), I² 65%	Not supported
Immune-cell infiltration	ssGSEA: adjusted |ρ| ≤ 0.24 (NK 0.14); MCP-counter: endothelial ρ 0.42, cytotoxic and NK scores null	Weak / descriptive
Immune-checkpoint genes	Checkpoint genes: all Spearman FDR > 0.4; none significant after adjustment	Not supported
WGCNA context	In the stromal/EMT turquoise module; non-hub gene (kME rank 1,516 of 1,747 module genes)	Weak / descriptive
Promoter hypermethylation	8 promoter CpGs: window Δβ −0.03; adjusted ρ −0.06 (P 0.29); decision rule not met	Not supported
Copy-number loss / somatic mutation as cause	Deep deletion 1.6% (any loss 58.9%); GLP2R mutated in 2.2%; adjusted CNV ρ 0.00	Not supported
Mutational-burden link	ρ −0.29 falls to −0.13 after MSI/site adjustment (MSS-only −0.14)	Weak / descriptive
Prognostic signature	8-gene LASSO: held-out test C-index 0.518; GSE39582 0.548; GSE17536 0.545	Not supported